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Review

# Sand mining in the Mekong Delta: Extent and compounded impacts

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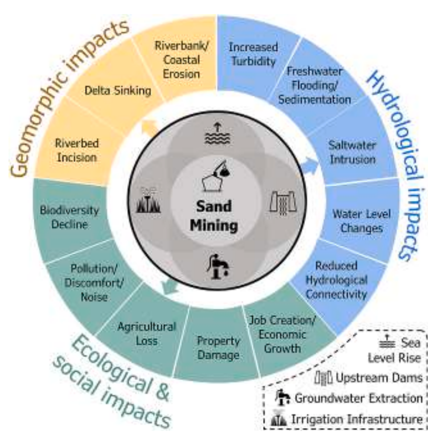
## HIGHLIGHTS

- The Mekong Delta is a critical case study for global sand mining issues.
- There are discrepancies in the volume of sand extracted from the riverbed.
- The ecological impacts of sand mining have not been well studied.
- The negative effects of sand mining have been amplified by other anthropogenic drivers.
- Remote sensing and deep learning can be used to map and quantify the extent of sand mining.

## GRAPHICAL ABSTRACT

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## ABSTRACT

Sand mining has accelerated in recent years primarily due to population increase and rapid urbanization. To meet demand, the rate of sand extraction often exceeds the rate of natural replenishment with serious environmental consequences. In this review paper, the Vietnamese Mekong Delta (VMD), a global hotspot for sand mining with a prolonged history of intensive riverbed extraction, is used as a representative case study to highlight the extent and compounded impacts of this activity. The sand mining budget of the VMD ranged from 8.5 to 45.7 $Mm^3$/yr. The large difference is due to the use of different methods to determine the sand mining budget as well as the difficulties associated with measuring the volume of sand extracted from the riverbed. Widespread illegal mining in the region further exacerbates the mismatch. The environmental consequences of riverbed sand mining include deeply incised riverbeds that result in riverbank and coastal erosion. Massive sediment removal has also led to river water level reductions, disrupted hydrological connectivity, and diminished floodplain inundation. In addition, the augmented backwater effect that results from riverbed lowering, amplifies the extent of saltwater intrusion in the dry season. While the physical and hydrological impacts of sand mining is well studied, studies on the ecological and socio-economic ramifications remain sparse. In addition, the ways in which upstream dams, irrigation infrastructure, excessive groundwater extraction, and sea-level rise (SLR) have amplified the effects of sand mining was also considered in this review. This paper concludes by

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advocating for the adoption of remote sensing-based approaches for effective mapping of sand mining activities and the need to mine sustainably to balance developmental needs with environmental conservation.

## 1. Introduction

Sand mining has been documented across the world from Asia (Chen et al., 2006; Gruel and Latrubesse, 2021; Mahadevan, 2019) to Africa (Bendixen et al., 2023; Chevallier, 2014; Mensah and Mattah, 2023) to America (Beiser, 2018b; da Silva et al., 2020; Sickmann et al., 2023). The construction industry is the largest user of sand as it is a key component of concrete. In addition, construction grade sand is also needed for land reclamation as the grains must be able to interlock. Meanwhile, silica sand is used for making glass, computer chips and in fracking for oil and gas (Beiser, 2018b). To meet demand, 32 to 50 billion tonnes of sand are extracted annually from the riverbed, seabed, beaches, and quarries (Beiser, 2018b; Bendixen MB et al., 2019). As the rate of sediment removal has often exceeded the pace of natural renewal, this has unfortunately resulted in serious environmental consequences (Beiser, 2018b; Best, 2019). To deal with the negative impacts of sand mining, the United Nations has issued resolutions on Mineral Resource Governance (cf. UNEA 4/19, UNEA 5/12) to improve the management of this critical mineral resource (UN Environment Programme, 2022a; UN Environment Programme, 2022b).

The Vietnamese Mekong Delta (VMD) is a regional sand mining hotspot in Southeast Asia and the extensive and excessive riverbed extraction has resulted in widespread environmental damage. The impacts of riverbed sand mining include riverbed incision which has contributed to bank instability. Unstable banks in turn threaten the safety of bridges and embankments (Brunier et al., 2014; Hackney et al., 2020). Riverbed sand mining has also decreased the frequency of floodplain flooding, depriving localities of ecosystem services such as the supply of fertile fluvial sediments and the flushing of contaminants (Hoang et al., 2018; Park et al., 2022). Moreover, the physical and hydrological changes from sand mining have worsened the extent of salinity intrusion in deltaic and coastal areas of the Mekong. The increase in soil salinities has meant that farmers are no longer able to grow salt-sensitive crops such as rice (Eslami et al., 2019; Loc et al., 2021a). From an ecological standpoint, the removal of sand from riverbeds has also displaced benthic fauna and flora, leading to a decline in biodiversity (Koehnken et al., 2020).

Upstream dams, irrigation infrastructure and excessive groundwater extraction have amplified some of the problems created by sand mining (Chua et al., 2022; Park et al., 2022). Upstream dams have impounded sediment in the reservoirs behind a dam and this has resulted in a sediment deficit as multiple dams along the length of the river have prevented a substantial volume of sediment from being transported downstream. The additional removal of sediment from the riverbed by sand mining activities have further increased the sediment deficit (Kondolf et al., 2014). Secondly, water withdrawal by irrigation infrastructure has reduced water levels in the river and together with riverbed incision from sand mining, there has been a drastic decline in the natural flooding in the Cambodian floodplains (Chua et al., 2022) and further downstream in the VMD (Park et al., 2020). A decline in flood frequency and extent is problematic as the ecological productivity of the Mekong is highly dependent on the annual flood pulse (Vu et al., 2021). Thirdly, excessive groundwater extraction in the lower Mekong has increased land subsidence and saltwater intrusion in the delta (Minderhoud et al., 2017). The removal of sand has made the VMD even more vulnerable to sea level rise. In future, climate change and rising sea levels may cause nearly half of the delta's land surface to be below the sea level with the remaining subaerial areas subject to pervasive salinity intrusion and flooding (Kondolf et al., 2018). Unfortunately, isolating the impacts associated with sand mining from other anthropogenic drivers is a complex process, which further complicates environmental impact assessments that need to be in place for sand mining activities.

The first step towards a systematic understanding of the environmental impacts that arise from sand mining is the quantification of the sand mining budget. Data on sand mining rates is scant in the VMD despite it being a sand mining hotspot. There is limited research because sand mining happens underwater, and in the case of the VMD, extensive bathymetric surveys over a large area and over multiple years are needed to estimate the changes in riverbed incision caused by sand mining activities (Gruel et al., 2022). However, the use of bathymetric difference may not always be applicable for other rivers as each river basin has a unique fluvial regime and land use activities that may not allow the sand mining budget to be estimated from solely bathymetric difference. Sand mining rates derived from government data are likely to be inaccurate as mining companies tend to under-report the amount of sand they have extracted (Bravard et al., 2013). The prevalence of illegal mining further complicate efforts to track the volume of sand that was actually mined (Yuen et al., 2024). As the scant information on sand mining activities in the Mekong Delta is generally unreliable, we have little understanding of the scale of sand extraction along the length of the Mekong River. While the negative impacts of sand mining are relatively well documented (Kondolf, 1997; Pandey et al., 2023; Rentier and Cammeraat, 2022), the extent in which they have manifested in the VMD remains uncertain.

Herein, I provide a comprehensive review of the extent and the effects of sand mining in the VMD from published literature. Personal insights from extensive fieldwork in the VMD in the last few years were also included to supplement the published data. In the first section, I reveal the extent of sand mining along the length of the Mekong River by summarizing the results of past studies in the VMD. The second section details the environmental and socio-economic impacts that have resulted from widespread sand mining activities along the Mekong River. In addition, I also highlight how other anthropogenic pressures in the Mekong have exacerbated the effects of sand mining. In the last section, I highlight the urgent need for a more systematic way of determining the sand mining budget and the importance of mining sustainably in the VMD. This review paper is a valuable addition to the literature in two ways. *Firstly*, this paper uses the VMD as a case study for understanding the multifaceted impacts of sand mining globally as extensive research has been carried out to document the various environmental impacts. By synthesizing studies that have documented the environmental and socio-economic consequences of sand mining in the VMD, my work will serve as an essential resource for future research and policymaking within the region. *Secondly*, the findings from this paper can be used to provide background information to facilitate studies to assess the effects of sand mining in other mega-deltas around the world.

## 2. Extent of sand mining in the Mekong Delta

This section reviews the socio-economic context of sand trade in the region over the past three decades, to provide background on the nature of the demand for sand. Subsequently, the magnitude and the extent of sand extraction is discussed. The final part of this section highlights the prevalence of illegal sand mining which has complicated attempts to account for all the sand that was extracted.

### 2.1. History of sand mining in the Mekong Delta

The Mekong River has the tenth largest sediment discharge load in the world at 160 Mt./yr before any dams were built (Meade, 1996). Sand, clay and silt are typically transported down the river as bedload or suspended sediment and deposited in lateral and median bars that have built up over rocky islands along the length of the river from Stung Treng to the upper delta (Bravard et al., 2014). Some of the sediment also accumulates in deeper parts of the channel as discontinuous lenticular patches (Nguyen et al., 2000). The abundance of sediment on the riverbed has provided a rich supply of sand for sand mining and mining activities has been ongoing in the VMD since the early 1990s (Bravard et al., 2013) (Fig. 1b). In the VMD, sand is usually extracted from the riverbed by barges with crane (BC) and the mined sand is immediately transferred to a sand transport boat (STB) that is positioned next to the BC (Fig. 1c). The STB will then transport the sand to a designated storage site on land after it is filled (Fig. 1d, e). The sand is eventually transported to where it is needed by another STB or by a truck (Fig. 1f). Boats equipped with suction pumps (BP) are also used to extract sand from the riverbed (Fig. 1g) but compared to the BCs, BPs are less efficient. BCs can collect 300 $m^3$ of sand in three hours while a BP need a day to collect 300 $m^3$ of sand (Huy, 2017). The sand that is extracted from the riverbed is usually used to support the construction of new buildings and infrastructure across Vietnam (Tuyen, 2023). Besides domestic demand, the high demand of sand from neighbouring countries like Singapore has also incentivized miners to extract more sand for export (Yuen et al., 2024) (Fig. 1a).

Since 2005, licensing for sand mining activities has been decentralized from the Central to the provincial government. Before sand mining can be carried out in the VMD, two licenses are required - one for the exploration of sand deposits and another for mining operations. As part of the licensing process, an environmental impact assessment (EIA) or an Environmental Protection Plan (EPP) must be in place and once mining operations end, rehabilitation efforts should be implemented. However, regulations are often ignored, and despite non-compliance, mining licenses continue to be issued and renewed by the provincial governments (Schiappacasse et al., 2019). The sand mining license issued to each sand mining operator includes information on where the mining operator is allowed to mine sand and how much sand they are allowed to extract from the riverbed (Yuen et al., 2024). Even though the Mekong River flows through multiple provinces in the VMD (Fig. 1b), how much sand is allowed to be extracted and where sand mining is permitted is determined independently by each provincial government without any inter-provincial dialogue (Schiappacasse et al., 2019). As the Mekong is a transboundary river, this approach is problematic as a systematic and integrated approach is needed to manage the sand reserves across the delta. In addition, it is possible that the allowable rates of sand extraction may have been arbitrarily determined based on provincial demand. This is because a news article quoted the deputy head of the Southern Institute of Water Resources Research (SIWRR) who mentioned that the sand supply in the VMD has never been calculated as provincial officials did not have the technical and financial capacity to do so (Ngoc et al., 2023).

### 2.2. Sand mining rates in the Mekong Delta

The rise in regional demand and the accompanying socio-environmental impacts have spurred investigations of the mining budget. Existing research have largely focused on specific geographical area(s) at the provincial scale. Although such information is useful for

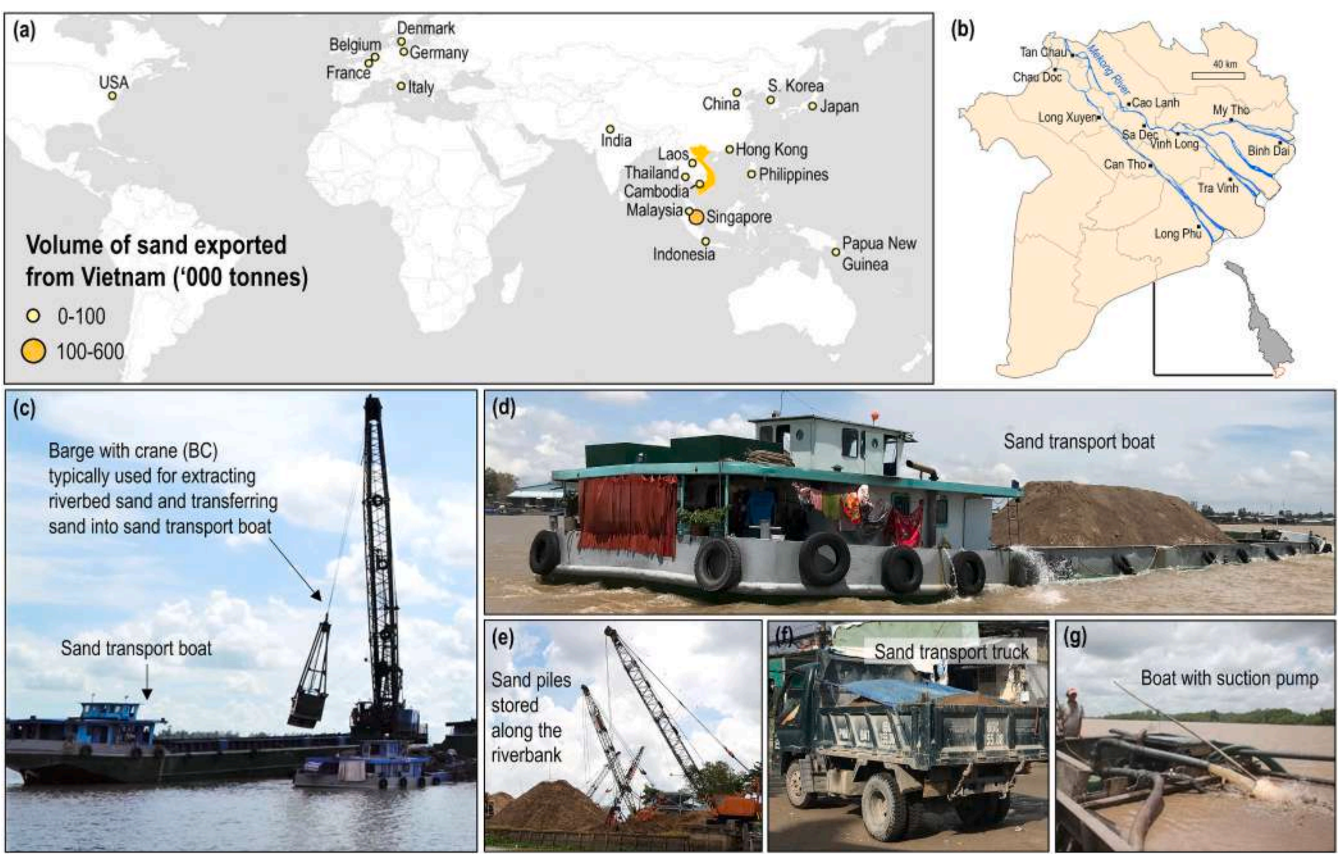


**Fig. 1.** Overview of regional sand demand from the Vietnamese Mekong Delta (VMD) and sand mining activities (photographs taken from field surveys). (a) Volume of sand exported from Vietnam to each country globally from 1990 to 2021. The sand export data was from the UN Comtrade Database (2022). (b) Study area map of the VMD. (c) Barges with crane (BC) are used to extract sand from the riverbed in the VMD. (d) Sand transport boats (STB) collect the sand scooped up from the riverbed by the BC and deliver the sand to a designated storage site on land. A STB includes a living quarter as workers may make multi-day voyages. This is evident from the presence of laundry (clothes, towels, and bedsheets). (e) Sand stored in piles along the riverbanks for future use. (f) Trucks are used to transport sand in the VMD. (g) Boats equipped with a suction pump (BP) are also used to extract sand from the riverbed. The photo of a boat with a suction pump was from Southern Institute of Water Resources Research (2013).

informing local regulations, they provide limited contribution to the understanding of the magnitude and impacts of sand mining across the entire VMD. Here, I synthesized previous research on sand mining in the VMD to provide a comparative analysis of the sand mining budget at the delta scale and provide an overview of the methods used to study sand mining in the VMD.

The first study that attempted to quantify the sand mining budget of the VMD was conducted by Brunier et al. (2014), where bathymetry data was used to show that 90 and 110 $Mm^3$ of sand was extracted from the Tien and Hau rivers over a 10-year period from 1998 to 2008. In that study, the average sand mining budget was 20 $Mm^3$/yr. When extrapolated to the scale of the VMD, 35.5 $Mm^3$/yr of sand was mined from the VMD annually. Another study by Vasilopoulos et al. (2021) over the same 10 year period revealed that 34.1 $Mm^3$/yr was extracted from the VMD from 1998 to 2008. In both studies, bathymetric data from 1998 to 2008 were obtained from the Mekong River Commission for analysis. Interest in quantifying the sand mining budget remained strong with Bravard et al. (2013) distributing questionnaire surveys to sand mining operators. From their survey, a sand mining budget of 7.75 $Mm^3$/yr for the year 2012 was derived, which was equivalent to a volume of 8.5 $Mm^3$/yr when scaled up to the size of the VMD. Following this, Eslami et al. (2019) used data collated from sand mining licenses and reported 28 $Mm^3$/yr of sand was removed from the VMD in 2015. Subsequently, Jordan et al. (2019) used data from local government departments to show that 17.77 $Mm^3$/yr of sand was mined from the VMD in 2018 (data from Ben Tre and Tien Giang provinces were excluded). In the second ten-year study period study by Vasilopoulos et al. (2021), 58.4 $Mm^3$/yr of sand was extracted from the VMD from 2008 to 2018 based on bathymetry data from 2008 to 2018 (Table 1).

As the early research on sand mining budgets used different methods that covered only a limited time and area, remote sensing methods that could cover a larger area over multiple years were then used to study sand mining. The first such study was by Hackney et al. (2021) who used five metre resolution PlanetScope imageries to estimate the sand mining budget along the Mekong River in Cambodia. The study period was from January 2016 to December 2020 and the sand extraction rates were estimated based on the storage capacity of sand transport boats. This value was multiplied by the number of sand transport boats and the number of calendar days in a month. Monthly values were then summed up to determine the volume of sand mined in a year. The next study that used remote sensing to study sand mining was by Gruel et al. (2022) who focused on the VMD. In that study, a bathymetry survey was conducted along a 100 km reach of the Tien River in July 2014 and September 2018 and a bathymetry difference map was generated to show the extent of accumulation and incision along the riverbed. Boats $\geq$70 m were manually identified, and a boat density map was then generated for 2014–2017. Corresponding values between bathymetry and boat density were plotted and the median bathymetry difference at each 0.1 boat density beam was calculated. A regression equation was then derived to estimate the riverbed incision rate based on daily boat density. The regression equation that was derived earlier was then applied to these four time periods (i.e. 2015–2017, 2016–2018, 2017–2019, 2018–2020) to generate an incision map for each three-year period. The volume of sand extracted/yr was derived from the incision rate/yr and channel area. The results revealed an increase in the VMD sand mining budget from 37.8 $Mm^3$/yr in 2015 to 47.3 $Mm^3$/yr in 2020 (Table 1). Among the provinces, Dong Thap showed the largest increase of the volume of sand mined, from 18.5 $Mm^3$/yr in 2014 to 25.7 $Mm^3$/yr in 2020.

Given that sand mining boats were manually identified in Gruel et al. (2022) and Hackney et al. (2021), deep learning frameworks were introduced to automate the detection of sand mining boats so that a large number of boats over a larger geographical area can be quickly identified. The first such project was by Smigaj et al. (2023) who used Faster R-CNN Resnet101 model and PlanetScope imagery to identify individual and clustered boats in the VMD. Although Smigaj et al. (2023) did not estimate the sand mining budget of the VMD, their method identified approximately 2180 vessels which was six times more than the 350 vessels identified in Gruel et al. (2022). One weakness of using PlanetScope imagery was the lack of data during cloudy conditions. As such, Kumar et al. (2024) used Sentinel-1 remote sensing products which do not have this limitation. A deep learning model based on the You Only Look Once version 5 (YOLOv5) DL algorithm was developed to automate the detection of sand mining boats from Sentinel-1 imageries. Thereafter, a BC boat density map was generated for 2014–2017 and corresponding values between bathymetry difference and BC boat density were plotted. The median bathymetry difference at each 0.1 BC density beam was calculated and a regression equation was derived. The regression model was applied between 2015 and 2022 in three-year intervals (i.e. 2015–2017, 2016–2018, …, 2020–2022) to estimate the BC density driven incision rate. Finally, the volume of sand

**Table 1**
A summary of sand mining studies in the VMD. The sand mining budget for each study were extrapolated to the scale of the delta with an area of 710 $km^2$ to get a rough estimate of the volume of sand mined in the VMD as many studies only surveyed a small area in the VMD. An area of 710 $km^2$ was used by Gruel et al. (2022) and we used the same value here.

| Reference | Study period | Study area | Size of study area ($km^2$) | Material and methods | Sand mining budget ($Mm^3$) | Sand mining budget of the VMD ($Mm^3$/yr) |
|---|---|---|---|---|---|---|
| Brunier et al. (2014) | 1998–2008 (10 years) | Tien & Hau rivers | 400 | Bathymetry data from Mekong River Commission for 1998 and 2008 | 200 (90 $Mm^3$ from the Tien River and 110 $Mm^3$ from the Hau River) | 35.5 |
| Vasilopoulos et al. (2021) | 1998–2008 (10 years) | Principal distributary channels of the VMD | 432 | Bathymetry data from Mekong River Commission for 1998 and 2008 | 341 | 34.1 |
| Bravard et al. (2013) | 2012 (1 year) | Tien & Hau rivers | 613 | Questionnaire surveys were given to contractors at each sand extraction site. | 7.75 | 8.5 |
| Eslami et al. (2019) | 2015 (1 year) | VMD | 710 | Issued sand mining licenses in Vietnam. | 28 | 28 |
| Jordan et al. (2019) | 2018 (1 year) | All VMD provinces except Ben Tre & Tien Giang | 572 | Data collected from three local government departments. | 17.77 | 22 |
| Vasilopoulos et al. (2021) | 2008–2018 (10 years) | Principal distributary channels of the VMD | 709 | Bathymetry data from Mekong River Commission for 1998 and 2008. A delta wide bathymetry survey was done in 2018. | 584 | 58.4 |
| Gruel et al. (2022) | 2015–2020 (6 years) | VMD | 710 | Bathymetry survey in 2014 and 2017, field survey in 2020 and 2021, boat density data from Sentinel-1 (2014–2017). | 253.58 | 42.3 |
| Kumar et al. (2024) | 2015–2022 (8 years) | VMD | 710 | Remote sensing based deep learning framework. | 365.86 | 45.73 |

extracted/yr was calculated by multiplying incision rate/yr with channel area. The use of this novel methodology revealed that 365.86 $Mm^3$ of sand was extracted from 2015 to 2022 which was equivalent to an annual sand extraction rate of 45.73 $Mm^3$/yr (Table 1).

### 2.3. Extent of illegal sand mining, a fundamental global challenge

The volume of sand extraction reported from observations in Gruel et al. (2022) was substantially higher than what was reported from studies based on government documents, pointing to the presence of illegal mining. Empirical observations also revealed that sand mining occurred in places where mining activities were banned. For example, although sand mining was banned in Ben Tre and Tien Giang provinces (Jordan et al., 2019), according to Gruel et al. (2022), 1.57 $Mm^3$/yr of sand was extracted in Ben Tre and 2.42 $Mm^3$/yr of sand was extracted from Tieng Giang from 2015 to 2020. Although the number of sand mining barges significantly decreased in both provinces from 2015 to 2020, sand mining did not stop completely and the number of sand mining boats had increased after 2020 (Gruel et al., 2022).

The magnitude of illegal mining in the VMD was recently revealed by Yuen et al. (2024). The study decoupled illegally mined sand budget from the total mined budget in the VMD by comparing allowable sand extraction rate determined by the provincial governments in Vietnam with the spatially gridded sand mining budget by Gruel et al. (2022). For the first time, illegal sand mining hotspots were identified and mapped. Along the Tien River, illegal mining was common around Tan Chau city in the north, from Cao Lanh to Vinh Long cities and around Tra Vinh city. Along the Hau River, illegal mining occurred along the length of the river from Long Xuyen to Can Tho cities. Interestingly, the amount of illegally mined sand had decreased from 16.7 $Mm^3$ in 2013 to 15.5 $Mm^3$ in 2018–2020. In the same period, the volume of sand mined from the VMD had increased from 37.8 $Mm^3$ to 45.4 $Mm^3$ over the same period. This contradiction was due to the sudden increase in allowable rate of sand extraction from 11.5 $Mm^3$ to 17.7 $Mm^3$ (+35 %). As the illegal mining rate was defined as the difference between the allowable rate of sand extraction set by the government and the actual volume of sand mined, an increase in the allowable rate of sand extraction resulted in a reduction in the rate of illegal mining. Increases in allowable rates of sand extraction seem unusual since 6.18 $\pm$ 2.07 Mt./yr or 3.86 $\pm$ 1.29 $Mm^3$/yr (based on a river sand density of 1600 kg/$m^3$) of sand is estimated to enter the Mekong Delta annually (Hackney et al., 2020) and the sand replenishment rate should not differ by a substantial amount inter-annually.

In August 2023, the Vietnamese police announced that a large illegal mining ring was busted in An Giang. Specifically, Trung Hậu Investment Joint Stock Company had secretly extracted >4.7 $Mm^3$ of sand from the Tien River, three times more than the authorized limit of 1.5 $Mm^3$. Some officials from the Department of Natural Resources and Environment of An Giang province and the leaders of the offending company are now facing punishment for their involvement in illegal mining (Việt Nam News, 2023c). As a result of the crackdown, sand prices have skyrocketed to VND320,000 to 330,000 (US$13 to 14) per cubic meter, VND 60,000 to 70,000 higher (US$2.50 to 3.00) than before as the supply of sand is now lowered (An and Ngoc, 2023). It is likely that the recent arrests were just the tip of the iceberg as 16 to 17 $Mm^3$ of sand was illegally mined in the VMD (Yuen et al., 2024). With two major construction projects in the VMD (i.e. 110 km Can Tho-Ca Mau Expressway and 188 km Chau Doc-Can Tho-Soc Trang Expressway) (An, 2023; Ngoc, 2023), it remains to be seen how effective and persistent the clamp down on illegal mining may be as up to 90 $Mm^3$ of sand will be required from 2022 to 2025 for expressway and local road projects (Tuyen, 2023)

Beyond the VMD, illegal sand mining has been documented in Cambodia (Global Witness, 2010; Lamb et al., 2019), China (Chen et al., 2006; Duan et al., 2019), Malaysia (Ashraf et al., 2011), Myanmar (Lamb et al., 2019) and Thailand (Rujivanarom, 2023). In India, illegal mining is controlled by the "sand mafia" with the local authorities often on the side of the illegal miners (Mahadevan, 2019). Allocation of human and monetary resources for monitoring such activities are generally lacking in these countries, creating conditions ideal for illegal mining. It is challenging to quantify illegal mining in general, as it is difficult to determine if the sand was illicitly sourced once it leaves the extraction point (Magliocca et al., 2021). In addition, there is a lack of information defining allowable rates of sand extraction in each locale. Therefore, I would like to highlight that crucial challenges remain in assessing illegal mining in river catchments and deltas around the region and beyond. To address the lack of data associated with illegal mining, there is a clear need for systematic, empirical, and cost-effective approaches to measure sand mining activities that are independent of government data.

## 3. Impacts of sand mining in the Mekong Delta

The previous section showed a considerable gap in the mining budget from different studies which was possibly due to widespread illegal mining. I then highlighted the need for systematic and empirical measurements of the sand mining budget. The next section reviews the various environmental and socio-economic impacts that have resulted from sand mining activities. After that I explain how upstream dams, irrigation infrastructure, excessive groundwater extraction and sea level rise have amplified some of the problems created by sand mining. A flow diagram illustrating the environmental processes that have resulted from sand mining is provided in Fig. 2.

### 3.1. Physical impacts

#### 3.1.1. Riverbed incision

Riverbed sand mining results in changes in channel morphology. The removal of sediment from the riverbed creates an incision on the riverbed (Fig. 2). The mobile nature of multiple BCs mean that multiple sand mining pits of variable shapes and sizes have been found along the length of the Tien and Hau Rivers, with some pits >10 m deep (Brunier et al., 2014; Lau et al., 2023). When the riverbed is incised due to sand mining, a nick point is created on the upstream end of the pit. Over time, the nick point migrates upstream as head cutting occurs and the slope of the pit increases. At the same time, excessive sediment removal has caused the water in the river channel to become sediment starved and this "hungry water" erodes the riverbed downstream. The riverbed is lowered eventually as the riverbed incision propagates upstream, downstream and laterally (Kondolf, 1997) (Fig. 3b, c). Although some of the sand mining pits may be partially refilled, local refilling rates remain uncertain. One estimate by Jordan et al. (2019) suggest that local refilling rates of sand in selected mining sites in VMD ranged from six

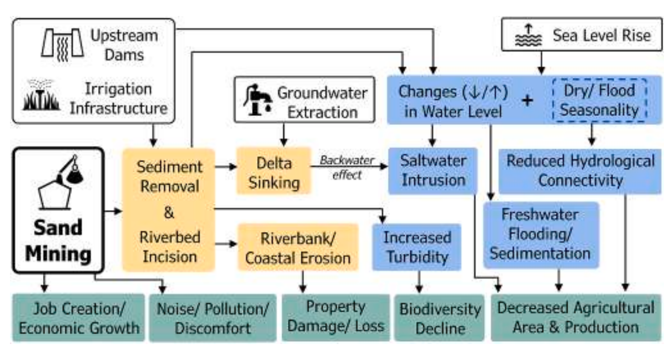


**Fig. 2.** Flow diagram showing the environmental processes and impacts that arise from sand mining in the Vietnamese Mekong Delta (VMD). Upstream dams, irrigation infrastructure and excessive groundwater extraction have exacerbated some of the problems created by sand mining. Sea level rise is considered a global driver that would amplify the effects of sand mining.

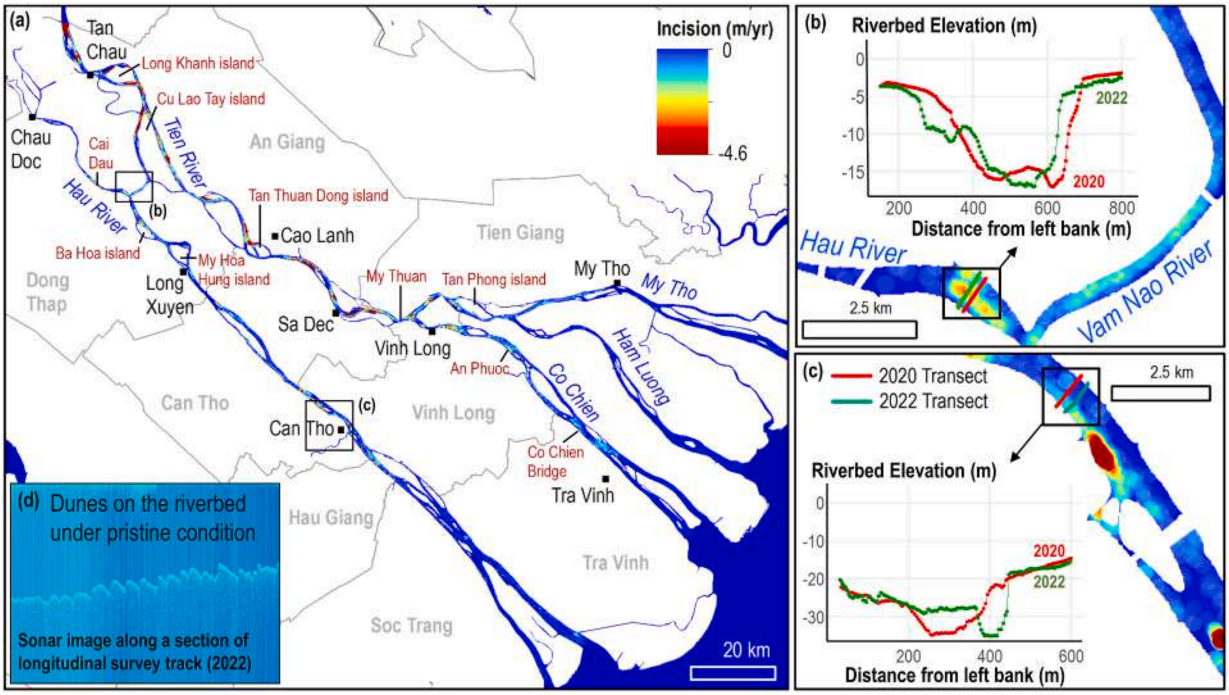


**Fig. 3.** Riverbed incisions in the Vietnamese Mekong Delta (VMD). (a) Sand mining hotspots identified by Binh et al. (2021), Brunier et al. (2014), Gruel et al. (2022) and Lau et al. (2023). (b) Enlarged view of sand mining hotspot downstream of Chau Doc city, with a comparison of the riverbed elevation in 2020 and 2022. (c) Enlarged view of sand mining hotspot near Can Tho city along the Hau River, with a comparison of the riverbed elevation in 2020 and 2022. Our comparison showed that sand mining created pockmarks on the riverbed and the riverbed was lowered over time. (d) Sonar image that showed dunes on the riverbed under pristine condition.

months to one year. In addition, riverbed sand mining also changes the grain size distribution of the sediment on the riverbed. This is because sand mining tends to target medium and coarse-grained sand and gravel on the riverbed, and the removal of this coarser armoured layer mean that the fine-grained sediment below may be exposed and mobilized (Koehnken et al., 2020).

Highly variable rates of riverbed incision have been reported based on diverse methods, study areas and time periods. A recent study by Lau et al. (2023) revealed that the upper reaches of the Tien and Hau river incised by 1.4 m (0.23 m/yr) from 2017 to 2022 and 1.3 m (0.33 m/yr) from 2017 to 2020 while the Co Chien branch of the Tien River experienced an incision rate of 0.7 m (0.18 m/yr) from 2017 to 2020. On the other hand, 0.2 m of sediment accumulated on the riverbed in the lower reaches of the Hau River from 2017 to 2020. In contrast, Binh et al. (2021)'s analysis of bathymetry data from 1998 to 2017 showed that the average incision rate was 0.16 m/yr on the upper Tien and Hau rivers. An intensification of sand mining activities had caused the rates of incision to increase to 0.5 m/yr from 2014 to 2017. At the delta scale, Gruel et al. (2022) highlighted that from 2015 to 2020, the Tien and the Hau Rivers experienced a total incision of 0.34 m (0.06 m/yr) as 42 $Mm^3$ of sand was extracted annually. The incision rate reported by Gruel et al. (2022) was slightly lower than the rate reported by Kumar et al. (2024) whose study suggested that the riverbed in the VMD had incised by 0.48 m (0.053 m/yr) from 2015 to 2022 as a result of the removal of nearly 366 $Mm^3$ of sand from the VMD. An overall analysis of channel depth data from 1998 to 2008 showed that the mean depth of the two rivers had increased by >0.13 m/yr as 20 $Mm^3$/yr of sediment was removed from the two rivers in 10 years. The mean thalweg depth also increased by 1.4 m with the thalweg of the Hau River less irregular than the Tien (Brunier et al., 2014). The incision rates reported in Brunier et al. (2014) in 1998–2008 differed from the incision rate reported by Vasilopoulos et al. (2021) by 0.04 m/yr as their study showed that the riverbed in the VMD had been lowered by 0.9 m (0.09 m/yr) from 1998 to 2008. In their second study period from 2008 to 2018, there was an increase in riverbed incision as the riverbed incised by 1.6 m (0.16 m/yr).

Despite highly variable channel incision rates, there was consistency among reported sites with significant channel incisions. Along the Tien River, locations with the lowest riverbed elevation in Lau et al. (2023) include (1) the meander after Sa Dec city, (2) after Long Khanh island near the beginning of the Tien River at Tan Chau, (3) around Tan Thuan Dong island near Cao Lanh city, (4) after Vinh Long city, and (5) near Tan Phong island on My Tho River. Likewise, the Tien River around Long Khanh island, Tan Thuan Dong Island, Sa Dec city, and Vinh Long city were also flagged as sand mining hotspots in Gruel et al. (2022). Additionally, bathymetric surveys by Binh et al. (2021) and Brunier et al. (2014) noted the presence of sand mining hotspots around Vam Nao River and Cu Lao Tay island along the Tien River. Deeply incised areas were also found along the Hau River, (1) near Can Tho city and (2) after My Hoa Hung island near Long Xuyen city (Lau et al., 2023). The area around Can Tho city was similarly recognized as a mining hotspot in Brunier et al. (2014) and Gruel et al. (2022) (Fig. 3, Table 2). Notably, sites with significant channel incision coincided with where illegal mining was taking place. Along the Tien River, the area from Sa Dec to Vinh Long city was an illegal sand mining hotspot with high rates of riverbed incision. The Tien River near the Cambodia-Vietnam border was also deeply incised as illegal mining was happening in the vicinity of Tan Chau city. Meanwhile, illegal sand extraction was rampant along parts of the Hau River from Long Xuyen to Can Tho cities (Yuen et al., 2024).

### 3.1.2. Bank erosion and collapse

Excessive sediment removal has caused the water in the river channel to become sediment starved and the "hungry water" have excess energy to erode riverbanks and induce bank collapse (Kondolf, 1997; Le et al., 2022). At the same time, the reduction in sediment load also cause sediment to be re-distributed locally (Jordan et al., 2019). The combination of "hungry water" and local sediment redistribution results in head cutting and erosion of the surrounding area. Head cutting and

**Table 2**
Summary of deeply incised channels/sand mining hotspots along the Tien and Hau Rivers from case studies in the VMD. The locations of several sand mining hotspots were annotated in Fig. 3a.

| Study period | Tien River | Hau River | Reference |
|---|---|---|---|
| 2017–2022 | • Meander after Sa Dec city<br>• After Long Khanh island<br>• Around Tan Thuan Dong island<br>• After Vinh Long city<br>• Near Tan Phong island<br>• Along the Cambodia-Vietnam border | • Near Can Tho city<br>• After My Hoa Hung island | Lau et al. (2023) |
| 2015–2020 | • Long Khanh island<br>• Channel heading towards Vam Nao<br>• Tan Thuan Dong island<br>• Around Sa Dec city<br>• Around the My Tho and Co Chien confluence<br>• Near Vinh Long city<br>• Around An Phuoc<br>• Downstream of Co Chien bridge | • Near Can Tho city<br>• Near Binh Hoa (Ba Hoa Island)<br>• Near Cai Dau | Gruel et al. (2022) |
| 1998–2017 | • Tan Chau to Cu Lao Tay island<br>• Vam Nao river<br>• 20 km upstream from My Thuan<br>• Sa Dec city | NA | Binh et al. (2021) |
| 1998–2008 | • Upstream of Ham Luong River<br>• Co Chien/Mekong bifurcations<br>• Vam Nao river<br>• East branch of Cu Lao Tay island | • Near Can Tho city | Brunier et al. (2014) |

erosion that result from an incised riverbed will cause the slope of the riverbed becomes steeper, and the river channel becomes wider over time (Figs. 2, 4a). Even the riverbanks a distance away from the sand mining sites are affected as well as the head cutting, and channel erosion propagates far beyond the original riverbed incision. Research that support the link between sand mining and bank erosion/collapse include Hackney et al. (2020) who pointed out that as little as 2 m of bed scour was enough to cause riverbanks to become seasonally unstable. Similarly, Binh et al. (2020c) suggested that scour holes on the riverbed were likely to be one of the main causes of riverbank erosion in the VMD.

Widespread riverbank erosion and bank collapse in the VMD (Fig. 4b) was confirmed by several case studies. For example, an analysis of Landsat satellite images from 2003 to 2017 by Jordan et al. (2019) revealed massive bank erosion within a 20 km stretch of the Tien River between Sa Dec city and the bifurcation spot of the Tien and Co Chien Rivers. Along this well-known sand mining hotspot, >1.86 km$^2$ of land was lost to erosion during the study period. Expanding on this study, analysis of Landsat imagery by Binh et al. (2020b) showed that the riverbank of a longer stretch of the Tien River from Tan Chau to My Thuan had a net erosion area of 1.5 km$^2$ from 2008 to 2014. An average of 2.64 m of the riverbank was eroded with some spots eroding by >60 m. The authorities are aware of the serious erosion problem as in 2023, the Vietnam Disaster and Dike Management Authority had reported that there were 585 major erosion hotspots in the VMD, spanning over 741 km. Out of the 585 erosion hotspots, 87 locations spanning 135 km were in danger of bank collapse (Hoang and Thu, 2023). Although there was no information on their proximity to sites of sand mining, sand mining was often blamed for causing erosion and bank collapse (Beiser, 2018a; Le, 2022; Le, 2023; Việt Nam News, 2023a). Lau et al. (2023)'s study confirmed the link between riverbank erosion and sand mining as multiple instances of bank erosion and bank collapse were identified near sites of lowered and irregular riverbeds. Unfortunately, in spite of efforts to reinforce the riverbanks (Fig. 4c), erosion has not abated (Ngoc et al., 2023) and instances of riverbank erosion and bank collapse that resulted in property losses (Fig. 4d) were often reported in the media. For example, between 2018 and 2020, a news article reported that 1808 houses in the Mekong Delta had collapsed into the river due to erosion

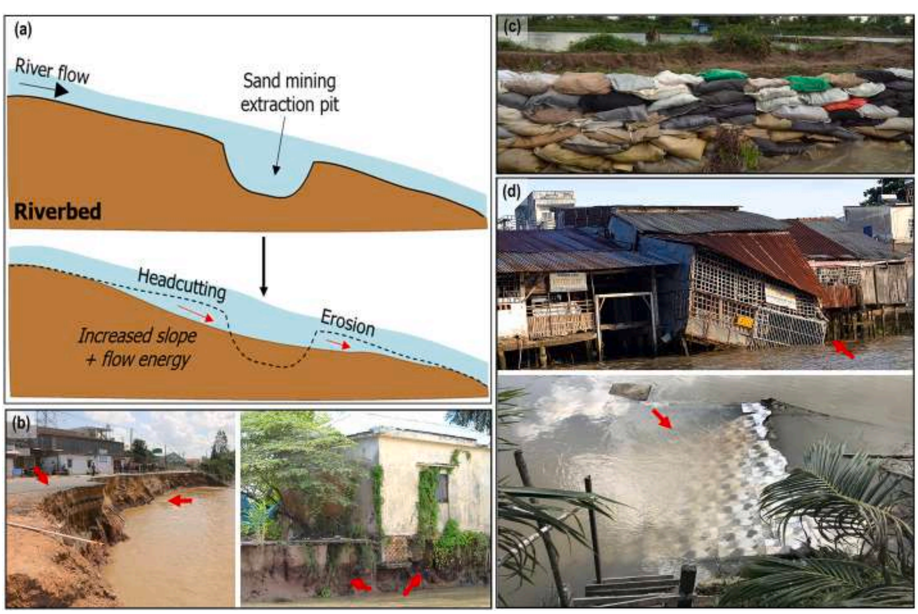


**Fig. 4.** Riverbank erosion in the Vietnamese Mekong Delta (VMD). (a) Head cutting and erosion that result from an incised riverbed will cause the slope of the riverbed to become steeper, and the river channel becomes wider over time. Eventually, the entire riverbank becomes destabilized. Even the riverbank a distance away from the sand mining sites are affected as well as the head cutting, and channel erosion propagates far beyond the original riverbed incision. (b) Severe erosion and bank collapse in the VMD. The photo on the left was taken from VietnamPlus (2019) while the photo on the right was from a field survey in June 2022. (c) Stacked sandbags were used to reinforce the riverbanks in Can Tho. (d) Collapsed houses were also observed along the river in Can Tho province.

(Le, 2022). Another news article pointed out that 2500 houses were swept away in An Giang, Dong Thap, Can Tho, Vinh Long and Ca Mau from 2018 to 2022 (Le, 2023). Farmlands, fishponds and shops were also lost when banks collapsed (Binh et al., 2021). In addition, excessive erosion had also destabilized the foundations of key infrastructure like roads, bridges and piers (Hackney et al., 2020).

While sand depletion resulting from sand mining made the riverbank prone to erosion and destabilization, the heterogenous nature of each river reach determined whether the riverbank was eroding or accreting. Khoi et al. (2020)'s study from 1989 to 2014 showed that the Tien and the Hau Rivers showed alternative trends of erosion and accretion even as sand mining was prevalent along the length of the two rivers. Along the Tien River, the left riverbanks in Tan Chau and Sa Dec eroded at a high rate of 30 m/yr and 42 m/yr respectively, while the right riverbank at Hong Ngu and Cao Lanh accreted at a high rate of between 18 and 45 m/yr. The high erosion and accretion were attributed to the sharp meander of the river which diverted streamflow towards the concave banks. High erosion and accretion rates were also observed on small islands along the length of the Mekong. In particular, the upstream areas around Long Khanh (Hong Ngu district), Cu Lao Tay (Thanh Binh district) and My Hoa Hung (Long Xuyen city) islands eroded at high rates of 6 to 14 m/yr and the downstream areas of these islands were accreting sediment at 8 to 15 m/yr. Meanwhile, Jordan et al. (2019) pointed out that the expansion of agricultural infrastructure resulted in 2.47 km$^2$ of accretion along a 20 km stretch of the Tien River with sand mining activities. As natural processes and/or human activities result in different rates of erosion or accretion along the length of the river, it is necessary to separate the river into smaller sections to allow differences in morphological and hydrological characteristics to be considered. At the same time, it is also important to evaluate these individual sections as part of the overall river system as the river catchment is hydrologically connected.

### 3.1.3. Coastal erosion

Sand mining has contributed to a substantial decline in the volume of sediment exported to the ocean from the Mekong River. This is highly problematic as river deltas are dependent on a sustained sediment supply to maintain the shoreline (Anthony et al., 2015). The reduced sediment supply makes the shoreline prone to coastal erosion and land loss as sediment removal is not balanced by deposits of new material along the shoreline. Historically, the VMD pro-graded seaward at a rate of up to >30 m/yr due to a plentiful supply of sediments that gets transported to the mouth of the river (Liu et al., 2017). The volume of sediment exported to the ocean from the Mekong River had decreased from 160 Mt./yr to 110 Mt./yr due to dam construction (Milliman and Farnswirth, 2011). A more recent estimate from 2012 to 2013 showed that only 40 Mt./yr of sediment made it to the ocean from the VMD (Nowacki et al., 2015). Out of this, the annual sand export from 2014 to 2015 was estimated to be 6.5 Mt./yr (Stephens et al., 2017). Excessive mining of sand from the riverbed has exacerbated the sediment deficit in the VMD and erosion by sediment starved water has turned the delta from "constructive progradation" to that of "destruction and erosion" (Kondolf, 1997; Li et al., 2017) (Fig. 2).

Multiple case studies have highlighted the extent of coastal erosion and shoreline change in the VMD. For example, Liu et al. (2017) found a reduction in the mean shoreline growth from 7.8 m/yr in 1973 to 2.8 m/yr in 2005. After 2005, shoreline growth was negative (−1.4 m/yr). Cumulatively, the growth of the deltaic land area had decreased from 4.3 km$^2$/yr (1973–1979) to 1.0 km$^2$/yr (1995–2005). Subsequently, a deficit of −0.05 km$^2$/yr was recorded in 2005–2015 which meant that the delta was in retreat. Erosion was particularly severe in the southeastern side of Ca Mau. Separately, Li et al. (2017) used Landsat satellite images to point out that only 44 % of the VMD shoreline displayed signs of erosion in 1973. However, by 2015, 66 % of the VMD shoreline showed signs of being eroded. A total of 10.8 km$^2$ of land was lost from 2005 to 2015 and erosion was particularly severe on the eastern side of the Ca Mau peninsula and on the northwestern side of the delta in the Gulf of Thailand (Fig. 5a). Over a similar time period, Besset et al. (2016)'s analysis of Landsat images from 1973 to 2014 revealed that nearly 70 % of the 160 km long South China Sea shoreline of the VMD showed signs of erosion. Coastal erosion was the most severe in Ca Mau province with land losses of between 20 and 50 m/yr in some places.

More recently, Anthony et al. (2015) used SPOT 5 images from 2003 to 2012 to show that 90 % of the 180 km long South China Sea coastline was in retreat with some parts receding by up to 50 m/yr. Although erosion was less severe along the lower energy Gulf of Thailand coast, 60 % of this 200 km long coast was affected by coastal erosion. Overall, the VMD had lost a total of 5 km$^2$ of coastal land between 2003 and 2012. Besides sand mining, shoreline orientation, limited southward redistribution of sediment deposited in front of the mouth of the Mekong River and the absence of sea dikes were important contributors to the more serious erosion on the southeastern side of Ca Mau (Besset et al.,

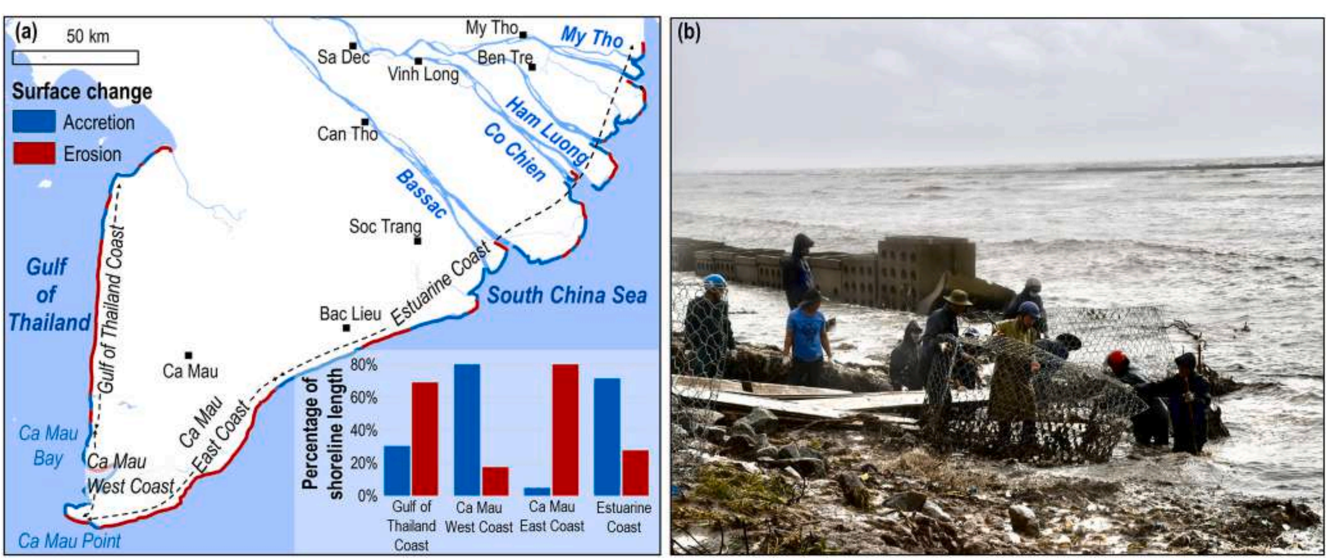


**Fig. 5.** (a) Extent of erosion and accretion along the VMD shoreline. The map was digitized from Li et al. (2017) and the study period was from 1973 to 2015. Shoreline along the Gulf of Thailand and in the southeastern part of Ca Mau were both flagged as coastal erosion hotspots. b) Eroded sea dykes in Ca Mau Peninsula, Vietnam. Photo credit: Nhân (2022).

2016). Another study blamed poor aquaculture pond construction, poor built quality of sea dikes and mangrove afforestation using only a single species for contributing to severe coastal erosion in Kien Giang province (Phong et al., 2017).

Coastal erosion in VMD was so severe that even sea dikes were damaged (although poor construction was a contributory factor) and land was lost (Fig. 5b). One news article pointed out that coastal erosion had damaged 2.7 km of coastal embankments in Ca Mau province (An, 2022a) while another news article stated that a 23 km section of sea dike in the same province was severely eroded (An, 2022b). In Hiep Thanh commune in Bac Lieu province, coastal erosion is consuming 30–100 m of farmland annually even as efforts were made to strengthen the natural embankments (Tran, 2018). Meanwhile, along the eastern and the southeastern coast of the VMD, rapid coastal erosion and mangrove degradation at many locations have left a narrow 100 m strip of forest (Phan et al., 2014). Overall, sand mining is a significant contributor to coastal erosion in the VMD, but attempting to quantify the extent in which sand mining has contributed to land loss would require the use of numerical models as apart from sand mining, sediment supply to the delta has been affected by upstream dams and irrigation infrastructure that have been built across the Mekong basin.

### *3.2. Hydrological impacts*

#### *3.2.1. Reduction in river-floodplain connectivity*

Sand extraction lowers the riverbed, leading to a decline in the water levels of the river channel (Fig. 2). The effects of riverbed incision on the hydrology of the VMD was investigated by Binh et al. (2021). Over two time periods (1980–1992 vs 1993–2018), the dry season discharge (March–June) of the Tien River had increased by up to 23 % but the dry season water level at My Thuan hydrological station had decreased by up to 46 % when the water level should increase as the discharged increased. Likewise, the Mekong's discharge showed a significant increase but the annual mean and monthly water levels at Tan Chau and My Thuan hydrological stations had decreased statistically in some months. According to (Binh et al., 2021), sand mining had partially contributed to an increase in riverbed incision from 0.16 m/yr in 1998–2014 to 0.5 m/yr in 2014–2017 and the lowering of the riverbed resulted in a lowering of water levels in the Mekong River.

The lower water levels have led to a reduction in hydrological connectivity (Fig. 2). A decrease in hydrological connectivity translates to a decrease in available water for irrigation and increased cost of pumping water for irrigation as the channel water level is now lower than the height of the river levee (Park, 2020; Park et al., 2020). In addition, a reduction in water levels in the Mekong has also resulted in reduced seasonal flooding. Annual flood frequency in the Long Xuyen Quadrangle (LXQ) decreased by 7.8 % from 36.7 % between 1995 and 2005 to 28.9 % between 2005 and 2015. The reduction in natural flooding was related partially to the lowering of the riverbed as rainfall trends around LXQ from 1984 to 2015 and daily river discharge at Kratie (300 km upstream) from 1995 to 2010 did not differ significantly. A disruption in flood regimes means a reduction in nutrient-rich flood sediments that nourish the soils (Chapman and Darby, 2016). Floodwaters can also remove contaminants, purify, and recharge aquifers, kill pests and mitigate saltwater intrusion (EEPSEA, 2011; Hoang et al., 2018). As flood frequency decreases, the frequency with which farmlands benefit from these environmental processes is reduced. A decline in flood frequency and extent is problematic as the ecological productivity of the Mekong is highly dependent on the annual flood pulse and any changes is likely to reduce inland fishery yields in the Mekong Delta (Vu et al., 2021).

Further upstream, sand mining activities along the length of the Mekong River in Cambodia has resulted in lower water levels and a reduction in the areal extent of flooding as well (Hackney et al., 2020; Hackney et al., 2021). Sand mining has been ongoing along the Lower Mekong River since the 1990s (Bravard et al., 2013) and 60 years of hydrological data from 1960 to 2019 showed that some parts of the floodplains are no longer flooded during the wet season as water levels fell. Moreover, the length of the flood season also decreased by 26 to 40 days and areas downstream had also experienced a greater delay in the arrival of floodwaters (Chua et al., 2022). Due to changes in the flood pulse, the annual reverse flow from the Mekong to the Tonle Sap Lake had decreased by 57 % from 48.7 $km^3$ in 1962–1972 to 31.7 $km^3$ in 2010–2018. Consequently, the wet season water levels at Tonle Sap Lake fell by 1.05 m in 2010–2019 compared to 1996–2009 and there was a 20.6 % reduction in the areal extent of the Ton le Sap Lake. Intensive sand mining along the Mekong River around Phnom Penh and from Phnom Penh to the Vietnamese border was one of the reasons behind falling water levels and a decrease in the areal extent of the Tonle Sap Lake (Bravard et al., 2013; Chua et al., 2022; Hackney et al., 2021). Similarly, Ng and Park (2021) also described how sand mining activities and the accompanying riverbed incision around Phnom Penh led to the shrinking of the Tonle Sap Lake. Hydrological data from 1980 to 2018 showed that the lake's volume had decreased as water levels showed a downward trend. No significant changes in stream discharge or rainfall were observed which suggest that riverbed incision due to intensive sand mining around Phnom Penh was accountable for the falling water levels. Given that there were also no sand mining activities in the Tonle Sap Lake and minimal mining along the Tonle Sap River (Ng and Park, 2021), the effect of sand mining in a distal location attest to the teleconnected nature of the river catchment.

#### *3.2.2. Intensifying salinity intrusion*

Although saltwater intrusion (SI) is a naturally occurring phenomena in the VMD, intensive riverbed sand mining has accentuated the extent of SI in the VMD (Fig. 2). Loc et al. (2021b) explained that sand mining along the Mekong had caused the water levels in the river to fall and this has resulted in increased backwater effect (BWE) and increased salinity intrusion (Fig. 6a). The results of their study revealed that daily water levels at Tan Chau and Chau Doc hydrological stations near the Cambodia-Vietnam border had decreased significantly since 1995 but the discharge trend remained unchanged. The inconsistent trend between water levels and discharge was partially attributed to the effect of riverbed incision caused by riverbed sand mining. Tidal amplification was evident when the mean annual water levels at three tidal stations along the coastline (Cau Quan on the Hau River, Son Doc and Huong My on the Tien River) showed a significantly increasing trend. Significant shifts of the tidal peaks in the coastal areas and the steepening slopes of the dry season rating curves developed from daily water levels of My Thuan and Can Tho stations suggest the presence of riverbed induced amplification of backwater effect.

This BWE was confirmed by Eslami et al. (2019) who used a numerical model to show that a 2 m riverbed incision in the Co Chien-Cung Hau channel may increase the extent of SI by 5 km and increase the salinity concentration by 1.5 PSU at a station that was 20 km from the sea. Along the Ham Luong River, a lowering of 20 % of the riverbed elevation was likely to increase SI length and salinity concentration 20 km from the sea by approximately 10 km and 2.5 PSU. These results confirm that riverbed incision results in lowered water levels in the upper VMD which in turn amplifies the BWE in the estuaries and reduce the water surface gradient between rivers and seas. Furthermore, riverbed incision also increases the tidal amplitude and leads to a reduction in river-sea gradient. This heightened tidal amplitude then accompanies the intrusion of saltwater body up the river, especially during the dry season (Fig. 6b). In the next 20 years, if the riverbed continues to be lowered, the tidal extent (TE) or the region of the delta where flow discharge is dominated by tidal forcing may increase by up to 56 km. This landward expansion of TE will exacerbate SI and create severe socio-economic impacts (Vasilopoulos et al., 2021).

SI limits agricultural productivity as it degrades the soil and salinizes freshwater resources (Tarolli et al., 2024) (Fig. 2). Notably, rice is unable to thrive in soils that have $>$4 g/L of salinity (Pham et al., 2018).

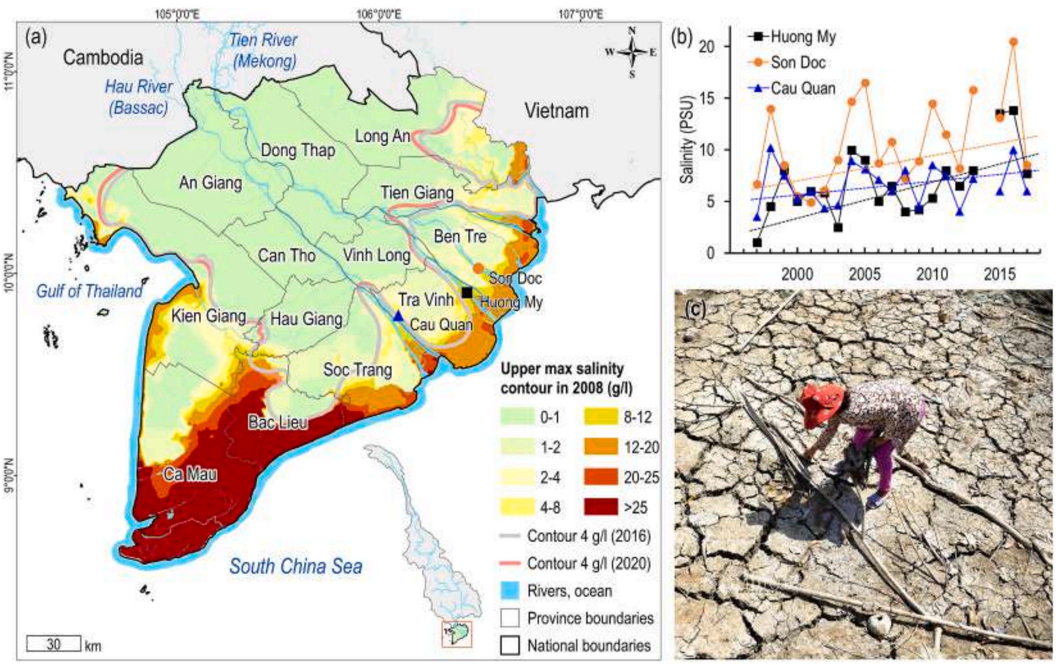


**Fig. 6.** (a) Intensifying saline intrusion in the VMD. The coloured areas represent the extent of SI across the VMD in a normal year (2008). Two 4 g/L isolines representing the two most recent drought years (2016 and 2020) were generated to illustrate the level of penetration of saline water into the inland areas. (b) Increasing trend of SI was indicated by the increase in annual mean salinity (PSU) measured at the surface water at the three major stations: Cau Quan (Hau River), Son Doc and Huong My (Tien River). The data that started from 1995 was from Eslami et al. (2019). (c) The riverbed had dried up in Tien Giang province due to the drought and the land was badly affected by the 2015–2016 salinity intrusion. Photo credit: Ba (2020).

Annually, approximately 1.8 million ha of the VMD is affected by dry-season salinity. Of this, 1.3 million ha is affected by saline water above 5 g/L (Smajgl et al., 2015). In addition, SI affects drinking water supplies. Consuming water high in sodium may lead to health problems such as hypertension, cardiovascular diseases, and skin issues (Rahman et al., 2019). In the dry season of 2019-2020, the VMD experienced a severe drought and saline water intruded up to 110 km inland, 11 km deeper as compared to the last drought event in 2015–2016. SI was also detected 2.5 to 3.5 months earlier than the annual average and lasted 30 days longer. Nearly 400,000 households had no access to fresh water while 30,000 ha of paddy rice in Ca Mau, 20,000 ha of fruit trees and 6500 ha of vegetables in Ben Tre were potentially damaged due to the water shortage (United Nations Viet Nam, 2020) (Fig. 6c). By 2030, simulation results revealed that SI will extend 10 to 15 km further by 2030. To adapt, farmers may have to adjust their crop calendars, store freshwater in the rainy season or change their production systems (Tran et al., 2024). Overall, SI has far-reaching consequences on food security, livelihoods and the health of the local community and the role of sand mining in contributing to SI should not be taken lightly. Although farmers can adapt and improvise to deal with changing conditions, the need to adapt suggest that the delta is no longer able to support the farmers in the way it used to.

### *3.3. Ecological impacts*

The Mekong Delta is a global biodiversity hotspot – a total of 470 species of fish were found in the Mekong. Out of this, 28 are endemic to the Mekong and four are only found in the delta. Notable mammals that live in the delta include the hairy-nosed otter, the Asian small-clawed otter and the dugong (Campbell, 2012). Another 247 bird species were identified in the Mekong with half of them wetland dependent and the remaining half migratory (Buckton and Safford, 2004). The Mekong River is also important for local food security as >1 million tonnes of fish are caught in the Cambodian and Vietnamese floodplains annually (Ziv et al., 2012). Despite its ecological significance and importance for local food security, there is limited research on how sand mining has affected biodiversity and ecosystem services in the VMD. Existing research on fisheries in the VMD have largely focused on the impacts of dams (Baird and Hogan, 2023; Ngor et al., 2018a; Stone, 2016), or the implications of overfishing and the use of illegal fishing methods (Ngor et al., 2018b; Vu et al., 2021). To fill this knowledge gap, I infer the ecological impacts of sand mining in the VMD from studies that focused on dams and fisheries and supplement my results with additional information from review papers on the effects of dredging on fisheries.

The removal of sand and gravel from riverbeds impact fisheries directly. Firstly, species living at sand mining sites get displaced or destroyed leading to biodiversity losses (Ahmed et al., 2020). Aquatic living resources on the bottom or near the bottom may also be removed (Karpova et al., 2021). Secondly, migratory routes may be disrupted as fish may not get to their traditional spawning sites due to changes in the aquatic landscape (Ahmed et al., 2020). Thirdly, the removal of sediment may also reduce nutrient loads. Related research on dams and sediment removal showed that under full dam buildout, the total transport of phosphorus (nutrient that controls primary production in freshwater ecosystem) was projected to reduce from 57 to 62 % to 47–53 % and induce a 12 to 27 % reduction in the primary productivity of aquatic systems in the Mekong Delta (Piman and Shrestha, 2017). Another study on dams showed that a 80 % reduction in the Mekong sediment load resulted in a 36 % decline of the total fish biomass in the Tonle Sap (Sarkkula and Koponen, 2010). Meanwhile, sediment plumes stirred up by sand mining activities increase water turbidity which may reduce visual acuity, inhibit foraging ability in benthic feeding species and damage gill tissues (Wenger et al., 2017). Increased water turbidity also reduces the amount of sunlight reaching aquatic vegetation and oxygen production may be suppressed. Overall, increased turbidity would lead to a biodiversity decline (Baran and Myschowoda, 2009) (Fig. 2). In addition, heavy metals released from mining activities may accumulate in gonad tissues and in eggshells causing developmental delays and larval deformities. On a separate note, underwater noise from dredging equipment may induce stress and damage auditory tissues

(Wenger et al., 2017).

Hydrological changes induced by sand mining activities may also affect fish. For example, riverbed sand mining was found to reduce flood extent and frequency (Park et al., 2020) which may negatively affect ecological productivity. Productivity is enhanced as floodwaters increase the size of aquatic environment available to fish and bring in nutrients to stimulate rapid growth of organisms in the food chain. Decaying matter and other contaminants are also removed by the floodwaters. Flood timing is critical to the continuity of life for some aquatic species because floodwaters provide hydrological cues to trigger seasonal migrations. When these cues become blurred or non-existent, fish species sensitive to these natural signals are not able to undertake their seasonal breeding migrations (Baran and Myschowoda, 2009). The importance of floods was reflected in how fish biomass and catch in the Tonle Sap Lake increased when floods were longer and more extensive (Halls and Hortle, 2021). In the Ton Le Sap system, the annual reversal of flow in the Ton Le Sap River is essential for ecosystem functioning. If the flow was not reversed or delayed, fish larvae from upstream spawning sites are unable to access the important floodplain habitats of the Tonle Sap system to complete their natural life cycle (Kang and Huang, 2022).

Understanding the ecological impacts of sand mining should be a priority in the VMD given its ecological richness. However, the exact contribution of sand mining to biodiversity decline is uncertain as pollution, overfishing and upstream dams also contribute to the decline of fisheries (Baran and Myschowoda, 2009; Chevalier et al., 2023). The impact of sand mining is likely to be significant as one report mentioned that the fish catch fell by 50 % after sand dredging started in Koh Kong province in Cambodia in 2008. As a result of sand mining, on some days fishermen caught no fish at all and even the crabs were gone (Boken, 2021; Global Witness, 2010). Another study also noted that fishermen had to fish further away from home if they were to catch anything (Lamb et al., 2019). More research is needed to understand the habitat ranges of key aquatic species including their spawning sites and spawning period so that sand mining activities can be adjusted accordingly. Research funding should also be allocated to assess the relationship between biological productivity and sediment loads, the bioavailability of contaminants released by sand mining and how individual species and age groups respond to sand mining activities in the vicinity. Lastly, the impact of sand mining on deep pools in the VMD can also be another area of research as these deep pools are known to provide refuge and shelters for many species (Mekong River Commission, 2009; Vu et al., 2009). Interestingly, the locations of deep pools along the Tien River in the VMD, seem to be near sand mining hotspots as sediments tend to accumulate in deeper parts of the channel (Nguyen et al., 2000). It is also possible that a lack of understanding of the location and ecological importance of deep pools (Vu et al., 2009) have contributed to the exploitation of sand from these critical habitats (Fig. 7a,b,c).

Apart from fish and other aquatic organisms, amphibians, birds and mammals are also affected by sand mining as they are also dependent on the Mekong River for food and shelter. Larson (2018) pointed out that amphibians such as the southern river terrapin in Southeast Asia, the northern river terrapin in Bangladesh and the gharial in India were threatened by sand mining activities. Meanwhile, falling water levels in Lake Poyang in China where intensive sand mining had been taking place for years have resulted in a worrisome decline in the number of juvenile cranes (Larson, 2018). Therefore, understanding how larger species respond to sand mining activities in the Mekong is another research priority to enable the implementation of appropriate conservation measures to protect vulnerable species.

### 3.4. Socio-economic impacts

The sand mining industry has created jobs such as dredgers who operate machinery to mine sand from the riverbed, pumpers who pump sand from sand barges to land and sand barge captains who sail along

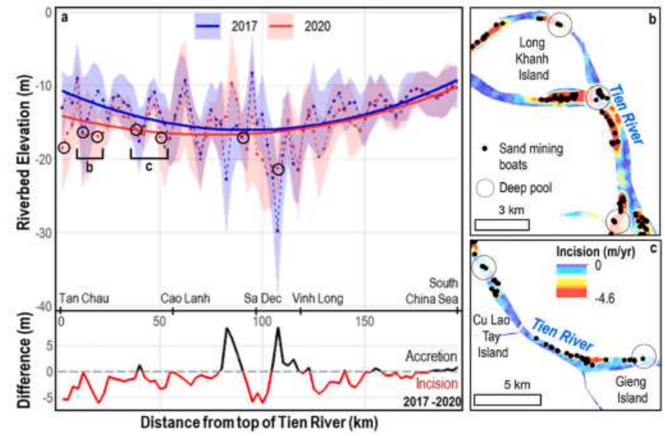


**Fig. 7.** Locations of deep pools along the Tien River of the Vietnamese Mekong Delta in relation to sand mining hotspots. Deep pool locations were obtained from a local ecological survey by the Mekong River Commission (2009). (a) Deep pool locations across a riverbed elevation longitudinal profile of the Tien River (Lau et al., 2023). Most of the deep pools coincided with recorded incisions between 2017 and 2020. (b) and (c) are enlarged views of deep pools that overlapped with the locations of sand mining boats that were recorded by Gruel et al. (2022).

the river transporting sand away from the sand extraction sites (Fig. 2). Such jobs require little, or no education and the work is less risky, less physically demanding and paid more than other menial jobs (Bendixen et al., 2023; Marschke et al., 2021). The sand mining companies also provide workers with three meals daily and a free housing (Lauzon, 2023). Sand mining also indirectly creates jobs in the construction industry as builders, drivers and sellers are needed. Overseas work opportunities may also be created in the construction industry of developed countries such as Singapore and these migrant construction workers may receive better remuneration compared to what they would get back home (Lamb et al., 2019). However, jobs in the sand industry or in construction usually require male members of the family to be away from home (Nhung, 2022; Tran, 2018). For example, those working on dredgers may have to live on a platform far from shore for prolonged periods while sand barge captains may have to make multi-day voyages from the sand extraction site to the sand collection point (Fig. 1d).

Sand mining has facilitated the development of housing and infrastructure projects that has increased urban development and allowed Vietnam to forge a high-income economy (World Bank Group, 2020) (Fig. 2). The increase in urbanization and development was reflected in Vietnam's GDP figures. In 2022, the GDP of Vietnam grew by 8.02 % as the industry and construction sector grew by 7.78 % (General Statistics Office of Vietnam, 2023). In addition, the development of housing and infrastructure projects also benefit the local population as people do not need to live in slums and require less time to travel from one place to another (Bendixen et al., 2023). Besides construction work, other jobs are also created as vendors, cleaners and security guards are needed in urban areas (Nhung, 2022). Sand mining activities also allow local governments to collect tax revenue if they are organized and legally carried out (Bendixen et al., 2023). This is a lucrative source of income for local governments and the money can be used to fund social initiatives to improve the welfare of the local population.

Unfortunately despite the job opportunities, financial gains and urban development, excessive and unregulated sand mining has imposed a wide range of environmental and socioeconomic consequences upon the population of the VMD. Firstly, the noise and dust generated from sand mining related activities are a another source of irritant for residents (Fig. 2). Li et al. (n.d.) surveyed 35 respondents in Can Tho who lived/worked near sand mining related activities to explore their perceptions towards sand mining. 74 % of the respondents felt that areas with sand mining-related activities were noisy. The noise originated from passing sand trucks, hydraulic equipment used to move the sand and from cement making. In addition, 89 % of respondents felt that the air quality had worsened because of sand mining related activities. Passing sand mining trucks also emitted exhaust while uncovered sand on the trucks would often fly out. One respondent complained about having to sweep the floor more often while another mentioned that he had been "eating sand" as the sand particles often got into his food. Although dust has the potential to irritate lungs and skin and noise have the potential to reduce the quality of life, little research has been done to look at the impact of mining on human health in the VMD and beyond (Bendixen et al., 2023).

During the survey, Li et al. (n.d.) also observed that roads with sand trucks traversing on it frequently had multiple potholes (Fig. 8c). Even though riverbed sand mining has created multiple problems for the local community, people who knew about sand mining and its impacts were hesitant to speak out as they felt powerless (Rousseau and Marschke, 2023; Tran et al., 2023). In contrast, villagers affected by sand mining activities in the Red River Delta have fought back against sand mining vessels that appear in the river by shouting and making noise on the riverbank or throwing rocks at the sand mining boats. Other disillusioned villagers have turned to mass and social media to highlight the presence of illegal mining in their neighbourhood (Runeckles et al., 2023). While it is not known if similar protests have occurred in the VMD, there are probably people in the VMD who are unhappy with the present situation.

Sand mining has also contributed to widespread bank erosion and collapse in the VMD. The threat of property loss from a collapsed riverbank have caused residents who live along the riverbanks to live in fear and anxiety (Truong, 2022; Việt Nam News, 2023c). According to a news article, there were 585 major erosion hotspots in the VMD spanning over 741 km in 2023, and with a delta population of 17.4 million (General Statistics Office of Vietnam, 2023; Hoang and Thu, 2023), the number of people whose land and properties are in jeopardy is not small. When homes and land are lost, victims may receive a small sum of money as compensation and/or are encouraged to relocate to a safer location. However, some have refused to move because they find the new location to be unsuitable (Le, 2022; Truong, 2022). A survey of 218 erosion affected households in An Giang and Dong Thap province found that 64 % of the respondents had difficulties dealing with the effects of river erosion as they did not have enough money to resettle and start over in a new job (Tri et al., 2023). Even business owners were not spared despite spending large sums of money on building embankments to protect their business and properties (Hoang et al., 2023) (Fig. 2). Tran et al. (2023) surveyed 34 residents in Can Tho and Long Xuyen city who lived near sites of bank collapse to assess their perception towards sand mining (Fig. 8a). Only 41 % of respondents ($n = 14$) were aware that sand mining was contributing to or triggering bank erosion/collapse (Fig. 8b). Some respondents blamed strong waves from passing boats for the erosion as multiple boats travelled down the river daily. After all, the waves from the passing boats were more visible than sand mining activities that may take place a distance away or at night. To deal with livelihood and property loss, some VMD residents have migrated to the city or joined the sand trade as jobs in the sand mining industry and other low skilled jobs in the city have a low barrier to entry (Nhung, 2022; Tran, 2018).

Overall, the socio-economic impacts of sand mining is understudied in the VMD. For investigators who are not conversant with local languages, conducting ethnographic research in the VMD would require reliable collaborators. To understand how sand mining has affected people and livelihoods in the VMD, interviews, questionnaire surveys and/or focus group discussions must be conducted in the local language (Eg. Vietnamese). The services of at least one competent translator are needed to overcome the language barrier if the researcher is a non-native speaker. Similarly, the application of research permit may involve the filling up of forms and documents that may be written in a foreign language. As such, a trustworthy local collaborator would also be needed to manage the project if the primary investigator is not a local.

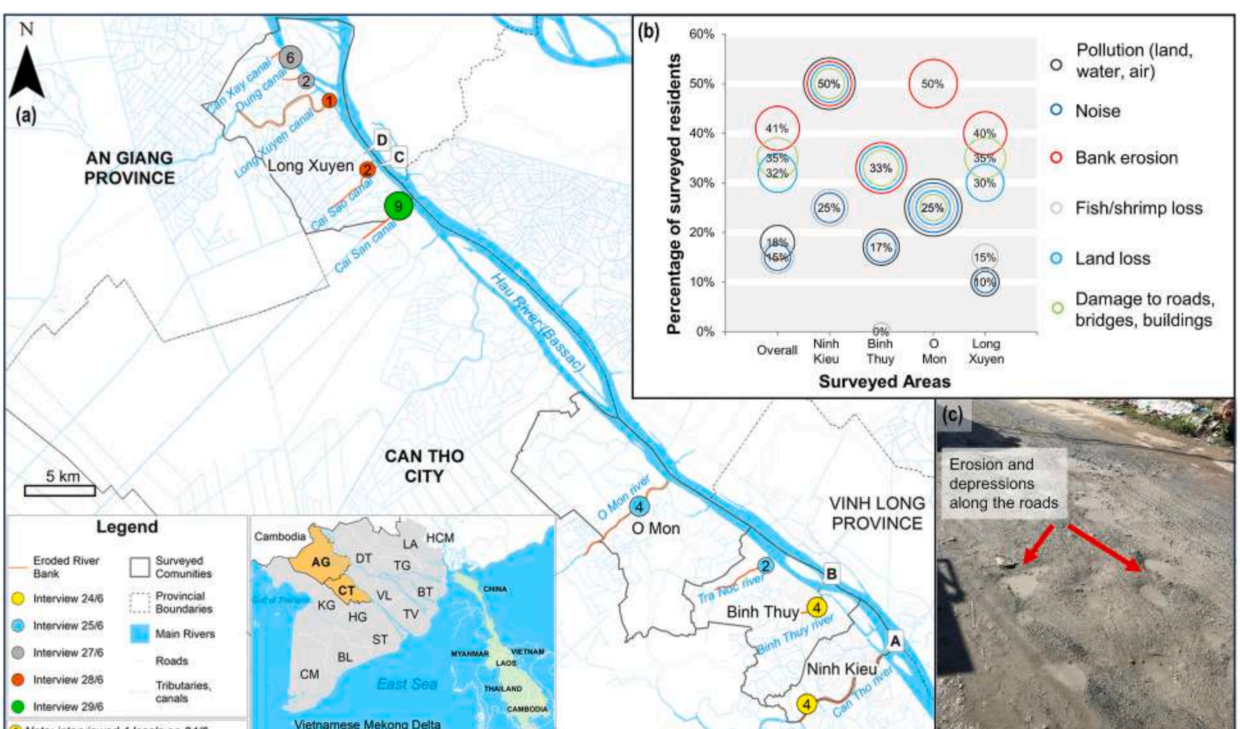


Fig. 8. Local community perception of sand mining impacts in the Vietnamese Mekong Delta (VMD). (a) Study area map of field surveys carried out in Long Xuyen and Can Tho province by Tran et al. (2023). A total of 34 Vietnamese locals were interviewed for the study. (b) Residents' perception on the various environmental consequences of sand mining activities. (c) A road heavily used by sand trucks had multiple potholes. Photo credit: Kai Wan Yuen.

Although socio-economic research is less straightforward, such research is still needed to improve our understanding of the socio-economic impacts of sand mining.

## 4. Compounding effects of multiple anthropogenic drivers

The tele-connected nature of the river catchment mean that sand mining and other anthropogenic drivers contribute to similar environmental impacts. Firstly, upstream dams have significantly reduced sediment fluxes (Fig. 2). The 11 mainstream dams in the Upper Mekong (Mekong River Commission, 2023) has trapped 598 Mt. of sediment since 2003 resulting in a 84 % reduction in sediment load at Chiang Saen, the hydrological station nearest to the Chinese dams (Chua and Lu, 2022). Further downstream, 89 hydropower dams have been commissioned in the Lower Mekong, including two in the Mekong mainstream (Mekong River Commission, 2009). As a result, sediment concentration at Kratie, before the Mekong flowed into the Cambodian floodplain and the VMD, decreased from 78 ± 22 Mt./yr (1995–2009) to 60 ± 21 My/yr (2010–2009) (Chua and Lu, 2022). In future, if all 133 planned dams were to be built, only 4 % of the pre-dam sediment load will reach the VMD (Kondolf et al., 2014).

Sand mining further exacerbates the extent of sediment depletion in the Mekong as an additional 38–48 Mm$^{3/}$/yr of sand is removed from the VMD (Gruel et al., 2022) (Fig. 2). Multiple studies have recognized how the removal of sediments by upstream dams and sand mining have resulted in sediment-starved hungry water that make the riverbanks and coastline vulnerable to erosion (Brunier et al., 2014; Schmitt et al., 2017). Riverbank collapse (Hackney et al., 2020), heavy rain (Darby et al., 2016) and land use changes (Chua and Lu, 2022) have the potential to mobilize and add sediments to the river network. But no prior research has been done to estimate the volume of this terrestrial sediment supply. Water levels in the river may also be lowered as the riverbed becomes incised with trickle down effects on hydrological connectivity, floodplain flooding and saltwater intrusion (Chua et al., 2022; Park et al., 2020; Park et al., 2022). However, as dams can release water to increase dry-season discharge or impound water to reduce wet-season discharge, the effect of dams on water levels can be uncertain (Binh et al., 2020a; Binh et al., 2020b) (Fig. 2). In addition, upstream dams have disrupted migratory routes and destroyed spawning and feeding grounds (Stone, 2016; Ziv et al., 2012). The result is an overall biodiversity decline in the Mekong (Sor et al., 2023).

Meanwhile, a dense network of irrigation canals have the potential to reduce sediment loads and modify water levels in the river to compound the hydrological changes brought on by dams and sand mining (Fig. 2). To explain why the southeastern part of the VMD was eroding at a rate of 2–3 km$^2$/yr, Tamura et al. (2020) suggested that the extensive network of irrigation infrastructure in Vietnam had resulted in a huge decline in coastal mud supply. Instead of being transported to the mouth of the Mekong, the sediment was either deposited in the floodplains when the floodwaters overflowed the dikes, or it ended up settling along the length of the canal. When large amounts of water are diverted to the crops, the water level in the river is lowered as well. There was a sharp decline in stream discharge after irrigation projects in Cambodia diverted the Mekong's flow. The decline in stream discharge, together with riverbed incision has led to a reduction in flood pulse in the Cambodian floodplains (Chua et al., 2022). In agreement, Erban and Gorelick (2016) found that the expansion of irrigated rice agriculture in Cambodia would mean that up to 31 % of the dry season Mekong River flow to Vietnam will be diverted. The withdrawal of such a large volume of water will result in significant reductions in the water supply in the VMD.

The dense network of dikes in the VMD that were built to keep floodwaters out to enable the cultivation of a third rice crop has also contributed to a reduction in flooding in the VMD (Fig. 2). While the effects of dike and riverbed mining on flood frequency cannot be decoupled due to their concurrent nature, the results of a multiple linear regression analysis using the mean annual water level at Chau Doc and dike coverage (both as explanatory variables) and annual flood frequency (response variable) revealed that water level variability contributed to flood frequency dynamics the most (~52 %) followed by dike construction (23 %) (Park et al., 2020). In short, irrigation canals and dikes have the potential to increase sediment removal and reduce water levels in the Mekong River. This would in turn accentuate the sediment deficit in the delta and further lower the water levels in the river creating trickle-down effects on hydrological connectivity, flood extent and saltwater intrusion.

Next, groundwater extraction is major cause of land subsidence in the VMD. Over-exploitation of ground water in the VMD has resulted in an average hydraulic head (groundwater level) decline of 30 cm/yr. The excessive groundwater extraction caused the delta to sink by an average of 1.6 cm/yr (Erban et al., 2014). Another study by Minderhoud et al. (2017) found that the average sinking rate in 2015 after 25 years of groundwater extraction in the VMD was 1.1 cm/yr (range: 0.7 to 1.8 cm/yr). Consequently, the coastal section of the VMD has been subsiding by 0.34 cm/yr to 1.13 cm/yr (Nguyen et al., 2023). The sea level around the VMD is also rising as the delta sinks. Sea level observations at tide gauges across Vietnam have recorded an average yearly increase of 3.3 mm from 1993 to 2014 (Hens et al., 2018) while the relative sea level rise along the VMD ranged from 0.56 cm/yr at Ben Trai to 1.35 cm/yr at My Thanh (Nguyen et al., 2023). Despite being a historically pro-grading landform, the delta is now eroding fast (Tamura et al., 2020) as sediment removal by upstream dams, sand mining activities and irrigation infrastructure has meant that less sediment is now available to sustain the delta. Given that much of the VMD is <2 m above sea level (Erban et al., 2014), groundwater-induced land subsidence coupled with sea level rise (SLR) will worsen the extent of saltwater intrusion (SI) in the VMD. With more areas affected by SI, agricultural productivity would also be affected as crops like rice are unable to thrive on salty soil (Fig. 2). These four factors of riverbed incision, land subsidence, upstream dams and SLR were previously identified by Loc et al. (2021a) as drivers of the intensification of SI across the VMD (Fig. 9). (Ba, 2020).

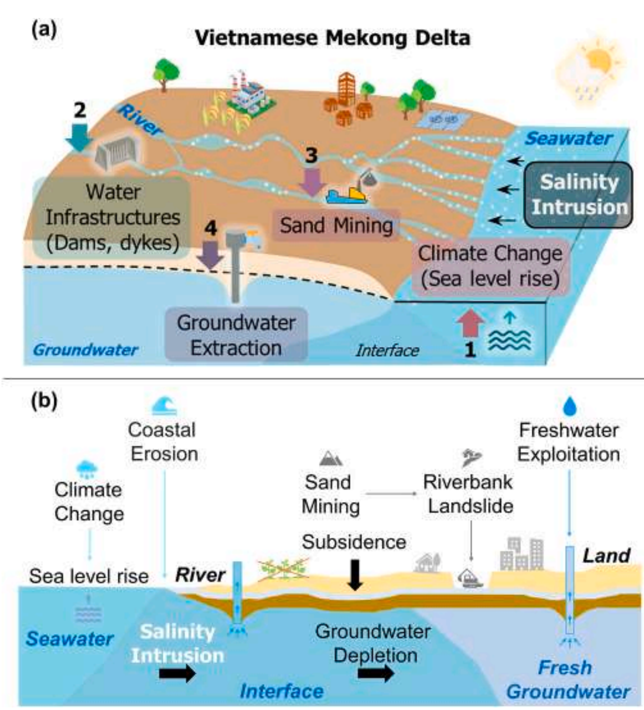


**Fig. 9.** A diagram showing how salinity intrusion in the Vietnamese Mekong Delta is due to four main environmental pressures: (1) Sea level rise, (2) irrigation infrastructure, (3) sand mining and (4) groundwater extraction.

## 5. Sustainable sand management in the Mekong and beyond

An essential part of improving the management of riverbed sand mining activities involves knowing where sand mining activities are taking place and accurately determining the volume of sand that had been extracted. Traditional fieldwork-based methods, such as in person observations and bathymetric surveys are effective and accurate but they are costly and logistically challenging in big river deltas as sand mining tend to take place in diffuse locations. A promising alternative lies in the use of remote sensing and artificial intelligence, especially machine learning. Remote sensing products allow sand mining activities to be documented over a large geographical area and data is available over a consistent period. Historical data can also be analysed to understand mining distribution over time. Research projects by Hackney et al. (2021) and Gruel et al. (2022) have highlighted the value of using remote sensing products to study and quantify sand mining activities along the length of the Mekong River. More recently, research by Smigaj et al. (2023) and Kumar et al. (2024) had demonstrated how the use of deep learning approaches can automate boat detection and allow a large number of images to be analysed efficiently. As compared to actual field observations, the combination of remote sensing and deep learning can significantly reduce costs as extensive fieldwork is not needed to survey boat traffic. This approach would be feasible for regions with limited resources.

Nevertheless, the use of remote sensing and deep learning approaches are not without limitations. The presence of cloud cover during the monsoon season has reduced the availability of usable PlanetScope data in Smigaj et al. (2023). In addition, the poor resolution of remote sensing products has sometimes complicated efforts to identify what was captured in the images. For example, images of vessels along the Solo River in Indonesia were grainy on Google Earth Pro making it difficult to tell if they were indeed sand mining boats. Even if the images are clear, images captured by remote sensing must be validated by an on-site survey that coincides with the overpassing of the satellite to confirm that that the vessel is indeed a sand mining boat. Observations of boat traffic along the Hau River in the VMD revealed that grain boats can easily be confused with sand transport boats as the colour of grain was close to the colour of sand (Fig. 10). Meanwhile, along the Mahakam River in Indonesia, coal was being transported in transport boats as coal was being mined in East Kalimantan (Bulolo, 2023). While contextual knowledge of coal mining activities was available in this case, it was possible for coal to be confused with black sand as both materials would look similar on top-down images captured by remote sensing sensors (Fig. 11g). On site surveys conducted as part of ground-truthing efforts would also allow the researchers to understand the different sand extraction methods that may be utilized in different locales. The use of different methods also mean that different sized boats may be deployed. In the VMD, barges with crane (BC) are used to extract sand from the riverbed and such boats were 37 ± 6 m long (Gruel et al., 2022). In the Irrawaddy Delta in Myanmar, smaller dredgers equipped with pumps were used to extract sand from the riverbed and such barges ranged from 40 to 70 m in length and had a surface area of 400–1200 m$^2$ (Gruel and Latrubesse, 2021). Over in Indonesia, sand mining in the Solo River is done on small boats with an iron scoop linked to a pulley or by sending divers to the bottom of the river to manually scoop sand. Sand may also be piped from the bottom of the riverbed and sucked to the surface with a diesel power engine similar to those used in the Irrawaddy (Sasmito, 2021). Overall, the different methods of sand extraction and the different dimensions of sand boats also mean that a deep learning model must be calibrated for each location to ensure accurate identification of the relevant sand mining boats (Fig. 11).

Next, remote sensing data should ideally be complemented by long term bathymetry data to ensure precise quantification of the sand mining budget. While Hackney et al. (2021) had pioneered using the dimensions of sand transport boats to estimate the volume of sand extracted from the Mekong River in Cambodia, some assumptions were made, such as using a uniform boat size, a fixed sand density and a standardized compaction factor, and assumed that each boat was fully loaded before being sent away. In the Mekong River in Vietnam, bathymetry data allowed sand mining sites to be confirmed and bed incision could be calculated (Gruel et al., 2022; Kumar et al., 2024). With more systematic monitoring approaches that can cover a larger area and provide consistent data over time, sand mining hotspots can be efficiently identified, and this information can be used to inform where bathymetry surveys should be carried out. In addition, sand mining hotspots determined from remote sensing products and bathymetry surveys can be compared against where sand mining licenses were issued to assess if sand mining was taking place in permitted mining areas. The volume of sand extracted can also be compared against the allowable rates of sand extraction to check that sand mining operators were mining within pre-defined limits.

Besides quantifying the distribution of sand mining activities and estimating the sand mining budget, there is also a need to know where the sand deposits are and what the sediment replenishment rates are relative to how much and where sand is needed. To inform policies on sustainable sand extraction, numerical models, based on field surveys are needed to map the deposition rates of river channels and identify channel areas where the natural replenishment rate is high enough for sand to be mined. This approach would allow sand mining to be carried out in a way that maintains geomorphologic and ecological balance. At present, such research is still at the ideation stage and extensive groundwork that include detailed environmental impact assessments and stakeholder participation are needed before data collection and policy implementation can take place. Ultimately, a delta wide approach is needed to manage sand mining activities in the VMD and other global deltas. As the Mekong and other major rivers flows through several


**Fig. 10.** Sand transport and grain boats along the Hau River in Chau Doc city, An Giang province in the VMD. To a casual observer, the colour of sand and grains appear quite similar and grain boats can be easily mistaken for transporting sand. From our fieldwork, there were some key differences to make distinguishing them easier. They include: (1) Sand transport boats are typically longer in length, (2) Sand piled on boats has an irregular shape and uneven texture while grains on a transport boat tend to be flatter and (3) Grain boats tend to have sacks on the sides of the boat and, the grain on the boat is sometimes covered. Photo credit: Kai Wan Yuen.

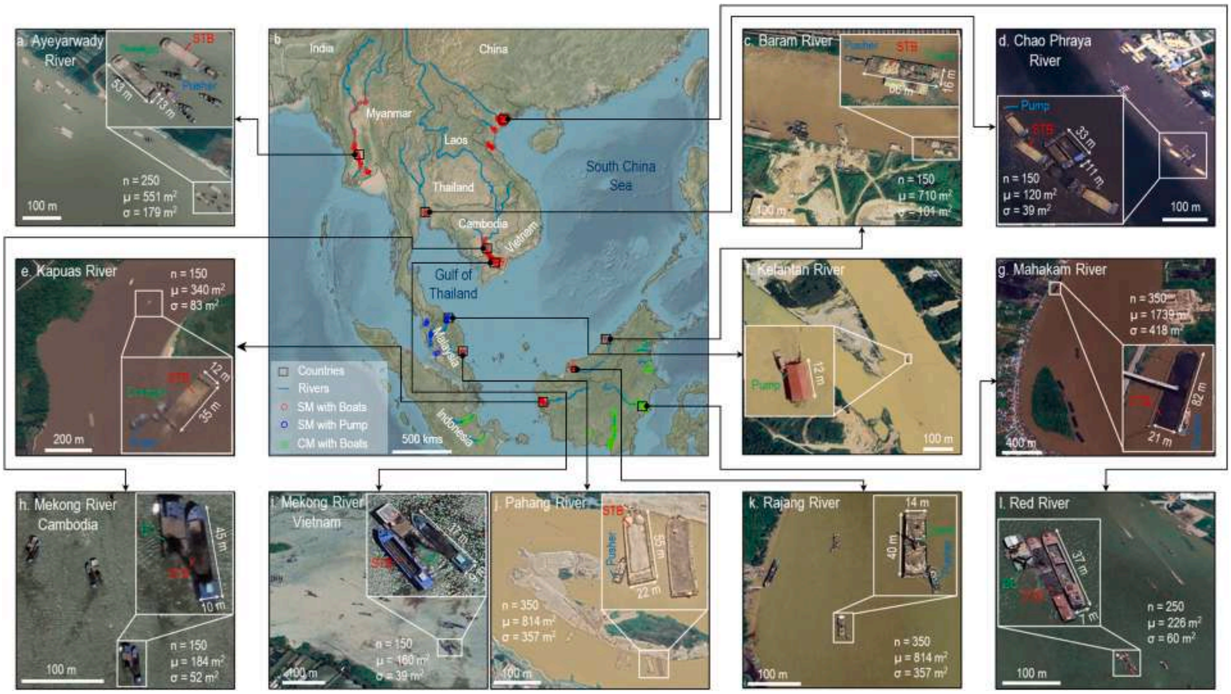


**Fig. 11.** Sand is mined from the riverbed in different ways across Southeast Asia. The boats used to extract and transport sand may also differ. Boats with dredging equipment that can extract a large volume of sand from the riverbed are common in the (a) Ayeyarwady River and (e) Kapuas River. Barges with crane (BC) that involve the use of a crane on a barge to extract sand from the riverbed can be found in (c) Baram River, (h) Mekong River in Cambodia, (i) Mekong River in Vietnam, (k) Rajang River and (l) Red River in Vietnam. Meanwhile, boats equipped with a small suction pump are used in the (d) Chao Phraya, (f) Kelantan River and (j) Pahang River. The sand holding capacity on a sand transport boat (STB) ranged from 160 $m^2$ in the Mekong River in Vietnam to 814 $m^2$ in the Pahang River. On the figure, 'n' indicates the sample size used to estimate the sand holding capacities of sand transport boats (STB). 'μ' denotes the mean value of the sand holding area, and 'σ' represents the standard deviation. Labels in green refer to how the sand is extracted from the riverbed, labels in red refer to boats used for transporting the material, while labels in blue refer to small boats that may be used to push or pull the transport boats. Coal transport boats (CTB) are common in the (g) Mahakam River, and such boats were highlighted in the figure to show that coal can be easily mistaken for black sand as both materials would look similar on top-down images captured by remote sensing sensors.

countries and multiple provinces within a country, a transboundary approach for managing the mining of sand is needed. Instead of a fragmentary approach by individual provincial governments, like what is currently happening in the VMD, sand reserves along a transboundary river should be managed at a regional scale that involve stakeholders in multiple provinces (and countries if possible). The demand for sand globally is projected to remain high and it is critical that this finite resource is not depleted too quickly with the help of interdisciplinary research and the involvement of multiple partners.

## 6. Summary and final remarks

Deltas are among the most productive and economically important ecosystems but they are also highly threatened by human activities (Day et al., 2016). The VMD is no exception, and I highlight the environmental consequences and socio-economic problems created by >30 years of riverbed sand mining in the VMD. The volume of sand extracted from the riverbed of the VMD ranged from 8.5 to 45.7 $Mm^3$/yr. The sand mining budget of the VMD is highly uncertain due to the use of different methods to determine the sand mining budget and the difficulties associated with measuring the volume of sand extracted from the riverbed. In addition, illegal mining makes it even harder to determine the exact volume of sand removed. The extensive sand mining across the VMD has left deep pockmarks in the riverbed leading to widespread bank erosion and collapse in the VMD. The removal of sediments has also contributed to coastal erosion as less sediment is transported to the coastal area to maintain the deltaic landform. Bank collapses often result in anxiety over loss of land and properties. Those affected by bank collapse are often forced to relocate to a safer location that may not be suitable for their needs and compensation is often insufficient for repairs or to start a new life elsewhere. Sand mining has hydrological consequences as well. For instance, riverbed incision results in lowered water levels in the river leading to reduced hydrological connectivity and floodplain flooding. In the dry season, the extent of saltwater intrusion has increased due to riverbed induced amplification of backwater effect, limiting the use of the land to only aquaculture, or growing of salt-tolerant crops. Meanwhile, the ecological consequences of sand mining are less well-studied with decline in fisheries often attributed to upstream dams, overfishing or illegal fishing methods. Further research is needed to examine the effects of sand mining on amphibians, birds, and mammals native to the VMD. On the socio-economic side, while sand mining has created jobs, facilitated urban development and economic growth in Vietnam, the people living in the VMD are left to bear the environmental and socio-economic consequences of (unmanaged) sand mining.

Ultimately, the impacts of sand mining are often compounded by upstream dams, irrigation infrastructure and excessive groundwater extraction. Multiple upstream dams have impounded a substantial volume of sediment upstream in their reservoirs and the sediment is not transported downstream to the VMD. Riverbed sand mining has arguably worsened the sediment deficit as whatever sediment that is left in the VMD is being removed from the riverbed to meet Vietnam's development needs or exported to other countries. Furthermore, irrigation infrastructure has also contributed to sediment removal. In this case, the sediment may be deposited in the fields, or it may end up settling along the length of the canal. In addition, the water levels in the river are also lowered when river water is diverted to the fields for irrigation with accompanying reductions on flood pulse. Another pertinent issue in the VMD is the excessive withdrawals of groundwater which has increased land subsidence and worsened saltwater intrusion as global sea levels rise.

Lastly, to alleviate the negative consequences of sand mining, I propose the development of a remote sensing-based monitoring system that can map sand mining activities and estimate the sand mining budget efficiently with only the minimum need for traditional field surveys that can be costly and logistically challenging given the size of large rivers. Rigorous scientific research should also be undertaken to identify fluvial reaches where sustainable sand extraction can be carried out to meet the high demand for sand in Vietnam. Allowable rates of sand mining should also be set, monitored, and enforced to ensure that the rate of sediment withdrawal does not exceed the rate of natural replenishment.

## CRediT authorship contribution statement

**Edward Park:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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